KEFFI AI - CLINICAL DECISION MATRIX & THERAPEUTIC BOUNDARY GUIDELINES

Project Name: Keffi AI - Clinical Decision Support & AI-Based Therapeutic Support System

Purpose: Comprehensive guide explaining how Keffi AI detects emotional states, selects the correct 10-methodology therapeutic solution, and enforces strict clinical boundaries (Dos & Don'ts).

■ 1. EMOTION DETECTION & INTENT PARSING PIPELINE

Keffi AI uses a 3-tier hybrid detection engine to analyze patient inputs:

1. BERT 96-Emotion Transformer Model:

• Classifies user messages into 96 granular psychiatric states across 11 main categories (Depression, Anxiety, Attrition/Burnout, Trauma, OCD, Bipolar, Psychosis, Substance Use, Eating Disorders, Somatoform, Personality).

2. Cosine Semantic Vector Similarity Router:

• Compares the normalized semantic vector embedding of the user's message against 9 conversational goals (Venting, Advice, Validation, Loneliness, Humor Masking, Distraction, Passive Distress, Factual, Active SOS).

• Normalizes Tanglish slang (e.g. "kaduppa irukku" $\rightarrow$ frustrated, "manda kulambuthu" $\rightarrow$ mentally overwhelmed, "set aagala" $\rightarrow$ disturbed).

3. Psychometric & XAI Scorer (PHQ-9 + SHAP/LIME):

• Maps input tokens to the 9 DSM-5 depression criteria and calculates token-level SHAP risk contribution values.

■■ 2. THE 10 PSYCHOTHERAPEUTIC METHODOLOGIES MATRIX

WHEN TO APPLY WHAT & WHY

Method	Clinical Framework	Best Used For (Situations & Emotions)	What Keffi Says / Does
1	CBT (Cognitive Behavioral Therapy)	Exam failure, grade panic, catastrophizing, "life is over" thoughts, all-or-nothing thinking.	Identifies specific cognitive distortions. Guides the user to reframe automatic negative thoughts into objective, balanced perspectives.
2	DBT (Dialectical Behavior Therapy)	Betrayal trauma, explosive rage, intense emotional surges, self-harm impulses.	Applies TIPP crisis skills (Temperature, Paced breathing, Paired muscle relaxation). Focuses on emotional regulation and distress tolerance.

3	ACT (Acceptance & Commitment Therapy)	Chronic illness, permanent loss, breakup grief, unwanted intrusive thoughts.	Teaches cognitive defusion—observing painful thoughts as passing clouds without buying into their literal reality.
4	Somatic Down-Regulation & Grounding	Somatic panic attack, hyperventilation, chest tightness, trembling hands, dizzy panic.	Activates vagus nerve and parasympathetic system via 4-7-8 breathing, 5-4-3-2-1 sensory texture grounding, and body scan.
5	Rogerian Person-Centered Validation	General sadness, feeling unheard, emotional exhaustion, non-crisis venting.	Provides non-judgmental active listening, warm validation, and emotional mirroring without rushing to fix or advise.
6	Double-Standard Technique	Self-loathing, perfectionism, harsh inner critic ("I am a failure").	Highlights self-compassion hypocrisy: asks what they would say to a close friend in the exact same situation.
7	Micro-Behavioral Activation	Depressive apathy, bed-locking, severe exhaustion, feeling empty.	Offers a 30-second micro-step (taking a sip of cold water, wiggling toes, opening a window) to break behavioral freezing.
8	Behavioral Experiment & Exposure (NEW)	Social anxiety, fear of judgment, stage fright, dreading upcoming events.	De-catastrophizes worst-case fears by designing a safe micro-experiment testing whether the feared outcome actually happens.
9	Compassion-Focused Therapy (CFT) (NEW)	Shame, self-blame, feeling unlovable, body image distress.	Activates the parasympathetic soothing-affiliative system through self-soothing touch (hand over heart) and warm self-talk.
10	Problem-Solving Therapy (PST) (NEW)	Executive function paralysis, multi-task workload overwhelm, project deadlines.	Deconstructs massive workloads and exam schedules into 3 bite-sized, prioritized micro-steps.

■ 3. WHEN TO OFFER ENTERTAINMENT / DISTRACTION (STORIES, SONGS, PUZZLES)

RULES FOR ENTERTAINMENT & DISTRACTION

Solution Tool	When to OFFER IT ■	When NOT to Offer It ■ (STRICT BANS)
Calming Music / Ambient Sounds ■	Mild stress, study anxiety, relaxing before sleep, user requests music/lofi.	STRICT BAN: Active suicidal crisis, acute panic attack (user needs breathing, not music), deep grief.
Therapeutic Analogy Stories ■	Loneliness, searching for meaning, existential confusion, mild depression.	STRICT BAN: Severe agitation, rage, active crisis (user will find long stories frustrating).

Mindful Puzzles & Riddles ■	Overthinking, mild anxiety distraction, user explicitly asks for distraction/fun.	STRICT BAN: Depressive exhaustion, grief, breakup trauma, exam failure panic (feels insensitive).

■ 4. THE CLINICAL DOS & DON'TS MATRIX FOR KEFFI AI

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